

# CS/EE 120B Custom Laboratory Project Report

## WPM Race

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### Final Video:

<https://youtube.com/shorts/UVTVYGR1OWo>

### Project Description

A typing test game, where the user utilizes a keyboard to quickly input a test phrase and is graded on accuracy and speed. High scores are saved to internal memory.

Test phrases are randomly generated each game, using a random seed set at initialization time.

### User Guide

Initialize the game by plugging it in

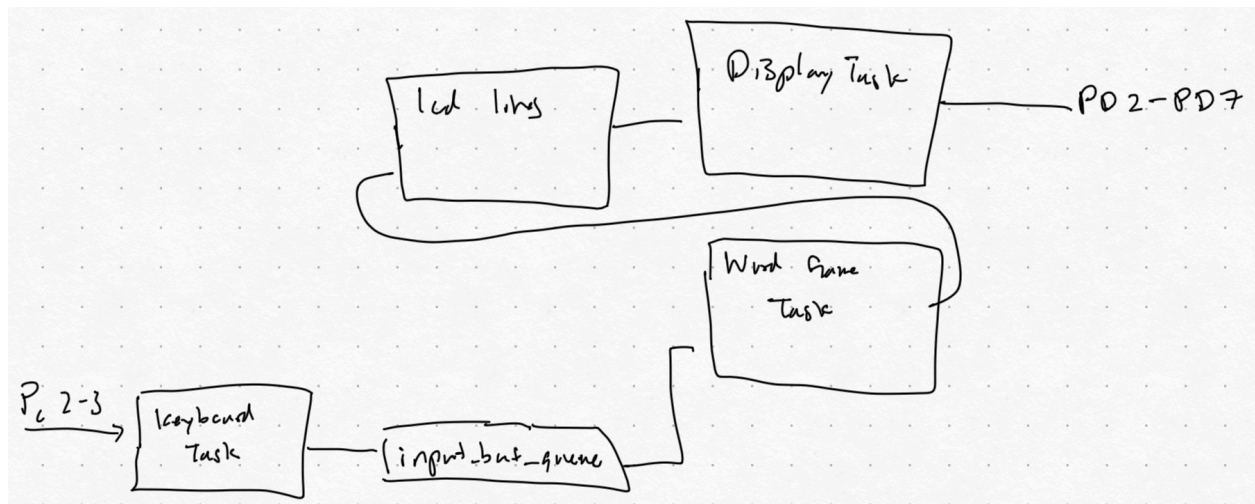
Enter to start the game; begin typing!

After 60 seconds, or after finishing the 255 character test string, the game ends!

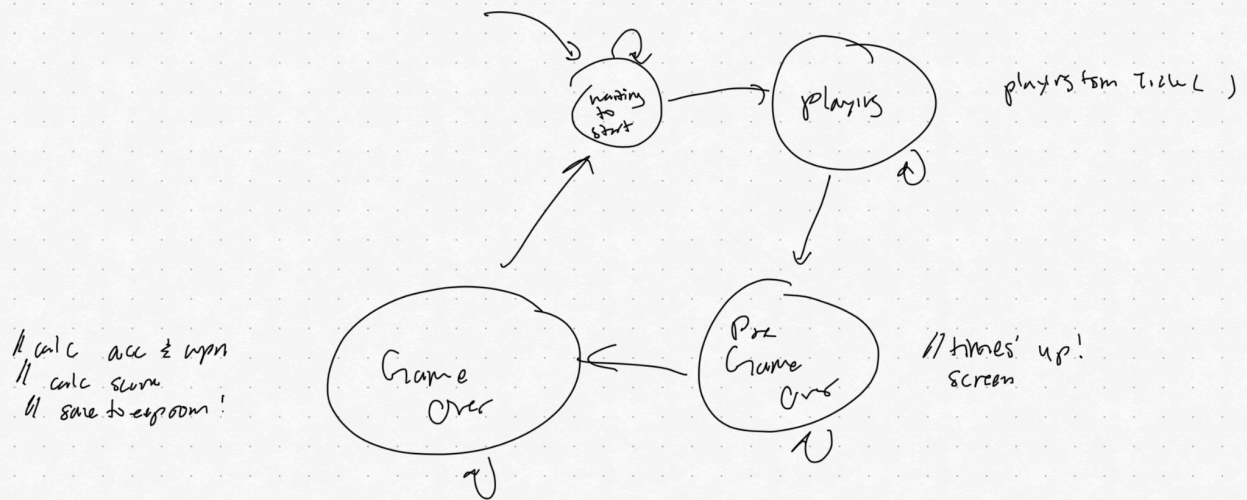
The game will then transition to a Times up! screen, and after pressing enter again, the game will tell you your WPM, accuracy, and see if you get a high score or not

### Task Diagram and SynchSM Diagrams

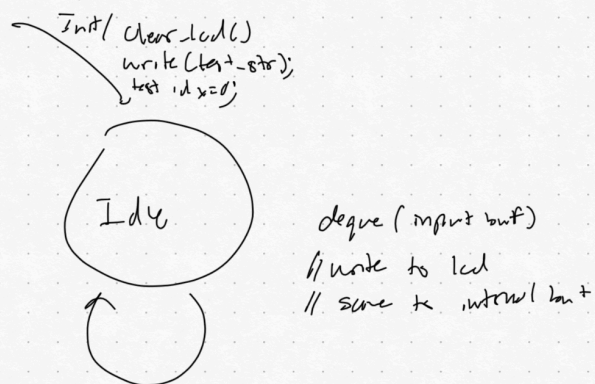
#### Task Diagram:



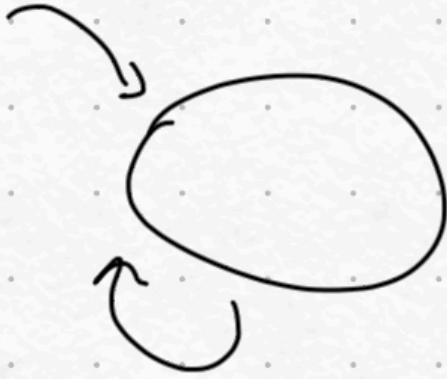
## Word Game Task



## Playing FSM Task



## Keyboard Task:



```
while (down (ps2 but))  
    enqueue (serial but, char)  
}
```

Keyboard task: 10ms

WordGameTask: 10ms

System GCD: 10ms

### Build Up on:

1. Flash + EEPROM Usage
  - a. EEPROM for highscore
  - b. Flash (PROGMEM) for huge string LUT for generating random strings
2. Keyboard
  - a. ps/2 protocol, scancode to char conversion
3. LCD Screen and Custom characters
  - a. Smiley and Frowny!
  - b. LCD used for displaying the typing words
4. Algorithms
  - a. Levenshtein string distance
  - b. XORShift32 RNG

### Known Shortcomings and/or Bugs

PS/2 keyboard sometimes ghosts keydowns, but i'm pretty sure part of it is due to the cheapness of the keyboard (using a USB adapter with my macbook also has the same issues)

### Acknowledgments

ps/2 spec- [https://oe7twj.at/images/6/6a/PS2\\_Keyboard.pdf](https://oe7twj.at/images/6/6a/PS2_Keyboard.pdf)

Levenshtein distance and xorshift32 algorithms- Wikipedia